# Heart rate, respiratory rate and airflow variability differences between stable and exacerbating COPD patients

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Shareable abstract (@ERSpublications)
**Physiological variability measures significantly differ between stable and exacerbating groups of COPD patients. This can be used to build exacerbation detection algorithms in the future, leading to improved patient outcomes.** https://bit.ly/4d2YMOC

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## Abstract

***Rationale*** Earlier identification and treatment of COPD exacerbations leads to improved clinical outcomes. Wearable technology has the ability to measure physiological signal variability, which is likely to be different in states of stability and exacerbation. The objectives of the present study were to analyse signals, including heart rate, respiratory rate and airflow, from a novel small wearable device, AcuPebble RE100 and compare differences in a group of stable and exacerbating participants.

***Methods*** Groups of stable and exacerbating adult participants with COPD were asked to wear AcuPebble RE100, which records physiological signals including heart rate, respiratory rate and airflow. Linear and nonlinear variability analysis was conducted on each of these time-series to detect differences between groups.

***Results*** A total of 51 participants (33 stable and 18 exacerbating) were analysed. Stable participants used the device for a median (interquartile range) of 18 nights (10–26). The exacerbating participants had significantly higher heart rate variability measures and a significantly lower heart rate complexity measure compared with stable participants. Respiratory rate variability and complexity were significantly increased in the exacerbating participants. Detrended fluctuation analysis demonstrated two crossover points in both populations, with the exacerbating participants demonstrating a significantly lower median $\alpha 3$ (0.50 (0.47–0.56) *versus* 0.69 (0.65–0.79); $p < 0.001$) compared with the stable population.

***Conclusion*** We have shown that significant differences exist in heart rate, respiratory rate and airflow variability measures between stable and exacerbating groups of COPD. This will help build exacerbation detection algorithms in the future.

## Introduction

Earlier identification and treatment of COPD exacerbations are likely to lead to improved patient morbidity and mortality while resulting in cost savings due to reduced healthcare utilisation and absenteeism [1, 2]. Changes in physiological signals such as heart rate (HR) and oxygen saturation have shown promise in this area, but simple linear analysis of these static signals has had limited clinical utility and failed to show a difference compared with symptom monitoring alone [3, 4].

Wearable technology has the potential to objectively measure continuous physiological signals which are likely to change from stable to exacerbating phases and highlight earlier, potentially subtle changes prior to symptom changes. A recent systematic review looking at the role of wearables to detect COPD exacerbations, found two previous studies that have shown a high positive predictive value (above 90%) in detecting an exacerbation. However, due to high attrition rates, one of these studies only included 13 participants in their analysis [4], and the other used a composite score relying on several parameters

including environmental sensors which are costly and difficult to replicate in real world settings [5]. Moreover, continuous measurement of physiological signals generates many datapoints, meaning novel analysis methods are needed to better understand this data. Previous work looking at oxygen saturation signal variability in a small number of COPD patients has shown promise in this area, but further work in this field is necessary [6].

The first step in creating an accurate system to detect physiological changes suggesting the start of an exacerbation, is to use a simple device that measures common physiological signals and identifies key differences between the stable and exacerbating phases of COPD. Therefore, the main objective of our work as to analyse signals, including HR, respiratory rate (RR) and airflow, from a novel small wearable device, AcuPebble RE100 and compare differences in a group of stable and exacerbating participants. This objective is important as it guides the identification of exacerbation event signatures using a simple device, which is crucial for the development of future patient monitoring systems aimed at early detection.

## Methods

This prospective cross-sectional study received ethical approval from the Health Research Authority in England (IRAS 247489; REC reference 19/NI/0194). The study was also prospectively registered with clinicaltrials.gov (NCT04495062).

### *Inclusion/exclusion criteria*

Adult participants with a diagnosis of COPD, made with an exposure history of >10 pack-year smoking history and spirometry showing a post-bronchodilator forced expiratory volume in 1 s ($FEV_1$) to forced vital capacity (FVC) ratio <0.7 were included in the study. Participants were excluded if they were not fluent in English, had an allergy to the wearable adhesive dressing, had an impairment, meaning that they would not manage to use the technology on their own, had an implantable device, had a known diagnosis of concurrent sleep-disordered breathing or needed ventilatory support. Participants were split into two groups: stable COPD participants, recruited from outpatient clinics with no increase in their usual respiratory symptom burden at the time of consent; and exacerbating COPD participants recruited from their hospital admission. COPD exacerbations were defined as per the Global Initiative for Chronic Obstructive Lung Disease (GOLD) guidelines, (an increase in breathless and/or cough and sputum production that has worsened in <14 days) [7]. We only included exacerbations that led to a hospital admission, defined as severe by the GOLD guidelines. We also included only participants that were admitted specifically for a COPD exacerbation.

### *Wearable device*

The wearable device used for this study was AcuPebble RE100. Under the scope of this study, AcuPebble was not used as a medical device, but as an acoustic monitor to acquire signals, for which it is CE marked. AcuPebble RE100 is a small circular device with a diameter of 2.9 cm that attaches to the neck *via* a disposable medical grade adhesive. It records physiological acoustic signals and algorithms convert these sounds into RR (breaths·min$^{-1}$) recorded every 2 s, HR (beats per min (bpm)) recorded every 2 s and airflow (normalised volt) recorded every 0.1 s.

### *Study protocol*

Following informed consent, baseline data, including demographics, smoking history, spirometry results and past medical, social and drug history were collected for both groups. Participants also completed two validated questionnaires: the modified Medical Research Council (mMRC) breathlessness score [8] and the COPD Airway Assessment Tool (CAAT) [9]. Participants were shown how to use the AcuPebble RE100, which connected *via* Bluetooth to a mobile phone. The stable COPD group were asked to use AcuPebble RE100 while they slept at night for up to 30 days; while the exacerbating group were given the device within 48 h of admission to hospital and asked to wear the device continuously until discharge, and following discharge nocturnally for up to 28 days. For the main analysis we compared readings between stable COPD participants and hospitalised exacerbating participants, with data capture in the first 48 h of their admission.

### *Analysis methodology*

When comparing readings between patients, a fixed time-series duration was necessary to avoid bias. A 6-h nocturnal time window was used for both exacerbating and stable patients (supplementary material). The average across all nocturnal stable COPD recordings was used to compare with the average initial nocturnal recordings (within 48 h of admission) from the exacerbating group. We also looked at the nocturnal recordings 5 days after discharge for the exacerbating group. Recordings shorter than a 6-h duration were not included in the final analysis. Data recordings that had ⩾15% of missing data in any of the time-series were also not included in the final analysis.

For analysis of the HR and RR time-series, linear and nonlinear analysis methods were used. Linear measurements included the mean HR and mean RR as well as SDNN (SD of the inter-beat interval of successive heart beats (ms)), cSDNN (SDNN corrected for mean HR) and SDBB (SD of the breath-to-breath interval (ms)).

Nonlinear measurements included Poincaré plot indices (SD1 and SD2) and sample entropy (SE). SD1 is a measure of short-term variability and for HR, usually reflects the effect of respiration on the cardiac cycle, whereas SD2 is a measure of longer-term variability [10, 11]. SE is a measure of the amount of information in a physiological signal, reflecting the engagement of physiological control. Higher SE values indicate a more-engaged system with increased information processing and increased complexity [12]. Multiscale entropy (MSE) of both the HR and RR were also computed. MSE is an extension of SE while "zooming out" and is representative of information processing at a lower resolution [13].

Given the high resolution of airflow data available, detrended fluctuation analysis (DFA) was used. DFA is an analytic method that provides useful information on the relationship between different segments of a physiological time-series at different scales of a recorded signal [14, 15]. Preliminary analysis showed the DFA curve had two crossover points, and this meant there was a short-term, intermediate term and long-term scaling exponent, $\alpha 1$, $\alpha 2$ and $\alpha 3$, respectively.

The analyses were computed using well-known methods previously described in the literature, using freely available coding algorithms in MATLAB processing software. For SE calculations, settings of m at 2 and r at 0.2 were used, and MSE was calculated over 10 scales.

A summary of the calculated measures can be found in table 1. Further information can also be found in the supplementary material.

For the stable COPD population, we assessed correlations between the time-series measures (table 1) and $FEV_1$, CAAT score and mMRC. For the exacerbating population, we assessed correlations between the time-series measures (table 1) and the admission National Early Warning Score 2 (NEWS2) [16], the C-reactive protein and the Rome COPD exacerbation severity classification [1].

We assessed usability and acceptability of AcuPebble RE100 *via* a standardised feedback questionnaire.

**TABLE 1** Summary of the linear and nonlinear analysis methods used in the study

| Measure | Definition |
|---|---|
| **HR measures** | |
| Mean HR, bpm | The average HR |
| SDNN, ms | The SD of the inter-beat interval of successive heart beats |
| cSDNN, ms | The SDNN corrected for the mean HR |
| SD1 HR, ms | A measure of short-term HR variability |
| SD2 HR, ms | A measure of longer-term HR variability |
| SEHR | A measure of signal HR complexity and richness |
| MSE HR | MSE of HR. This looks at signal complexity when the data are "zoomed out" at different scales |
| **RR measures** | |
| Mean RR, breaths·min$^{-1}$ | The average RR |
| SDBB, ms | The SD of the inter-breath interval of successive breaths |
| SD1 RR | A measure of short-term RR variability |
| SD2 RR | A measure of longer-term RR variability |
| SERR | A measure of RR signal complexity and richness |
| MSE RR | MSE analysis of RR. This looks at signal complexity when the data are "zoomed out" at different scales |
| **Detrended fluctuation analysis measures** | |
| $\alpha 1$ | Short-term scaling exponent (roughly 3 s) |
| $\alpha 2$ | Medium-term scaling exponent (roughly 30 s) |
| $\alpha 3$ | Longer-term scaling exponent (more than 30 s) |

HR: heart rate; bpm: beats per minute; MSE: multiscale entropy; RR: respiratory rate.

To give some context we have also included data from a historical non-COPD group in the results section. The methodology for this historical group can be found in the supplementary material.

### *Statistical analysis*

All statistical analysis was conducted using the software Statistical Package for the Social Sciences (SPSS v.29). Baseline demographics between groups was compared using chi-squared tests for ordinal and categorical data. Continuous data were checked for normality and analysed using the independent t-test for parametric data or the Mann–Whitney U-test for non-parametric data. MSE analysis at varying scales was compared using a two-way ANOVA. A p-value $\leqslant 0.05$ was considered to be statistically significant.

## Results

200 participants were screened, of which 114 met eligibility criteria, and 59 were consented for the study, 36 with stable COPD and 23 undergoing an exacerbation. Due to short recordings and subsequent ineligibility, a total of 33 stable COPD participants and 18 exacerbating participants were analysed (51 in total). The study flow diagram is illustrated in figure 1. There were no significant differences in participant demographics between groups (table 2) except that the participants undergoing an exacerbation had a significantly lower median (interquartile range (IQR)) body mass index (BMI) compared with the stable population (18.9 $kg.m^{-2}$ (17.7–24.5) *versus* 27.1 $kg.m^{-2}$ (21.0–31.8); p=0.002). The exacerbating population had significantly more severe COPD with worse historic lung function and an increased CAAT score at the time of recruitment (table 3).

Stable COPD participants used the device for a median (IQR) of 18 nights (10–26) and for a median (IQR) of 8.3 h (6.8–9.8) per night. Therefore, overall, there was a total of 338 of 519 (65%) nocturnal HR recordings, 492 of 519 (95%) RR recordings and 345 of 519 (66%) airflow recordings available from 33 stable COPD participants.

Exacerbating COPD participants' first valid nocturnal recording within 48 h of admission was used to compare with the stable COPD population. Following deletion of missing time-series data, 9 of 18 (505) HR recordings, 18 of 18 (100%) RR recordings and 12 of 18 (67%) airflow recordings were available.

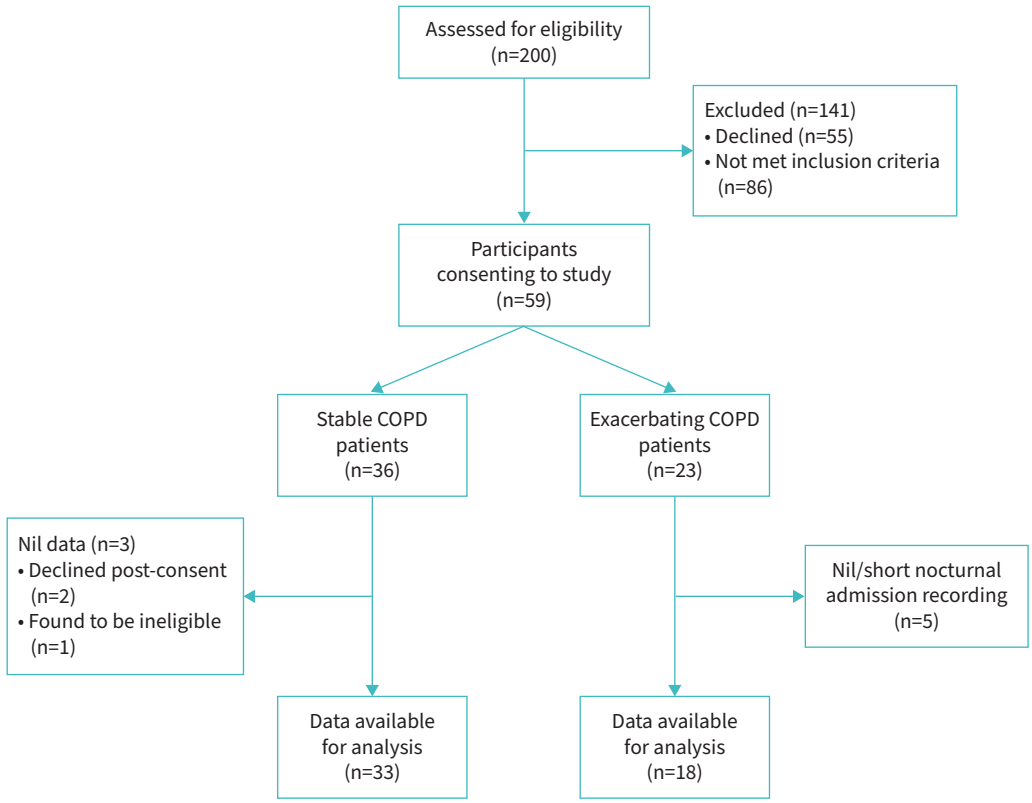


**FIGURE 1** Study flow diagram

**TABLE 2** Baseline characteristics of stable *versus* exacerbating participants

| Baseline characteristic | Stable COPD (n=33) | Exacerbating COPD (n=18) | p-value |
|---|---|---|---|
| **Male** | 20 (61) | 10 (56) | 0.73 |
| **Female** | 13 (39) | 8 (44) | 0.73 |
| **Age, years** | 67±10 | 64±9 | 0.46 |
| **Body mass index, kg·m$^{-2}$** | 27.1 (21.0–31.8) | 18.9 (17.7–24.5) | **0.002** |
| **Current smoker** | 6 (18) | 7 (39) | 0.11 |
| **Ex-smoker** | 27 (82) | 11 (61) | 0.11 |
| **Pack-year history** | 47 (25–64) | 43 (40–50) | 0.97 |
| **Mobility** | | | 0.69 |
| Independent | 20 (61) | 13 (72) | |
| Uses a stick | 11 (33) | 4 (22) | |
| Uses a frame | 2 (6) | 1 (6) | |
| **Independent with regards to activities of daily living** | 31 (94) | 18 (100) | 0.29 |
| **Medical comorbidities** | | | |
| α-1 Antitrypsin | 4 (12) | 0 | 0.12 |
| Atrial fibrillation | 3 (9) | 0 | 0.19 |
| Cerebrovascular disease | 0 | 0 | NA |
| Hypercholesterolaemia | 9 (27) | 5 (28) | 0.97 |
| Hypertension | 11 (33) | 5 (28) | 0.68 |
| Ischaemic heart disease | 5 (15) | 2 (11) | 0.69 |
| Oxygen therapy | 2 (6) | 1 (6) | 0.94 |
| Peripheral vascular disease | 1 (3) | 0 | 0.46 |
| Type 2 diabetes mellitus | 5 (15) | 1 (6) | 0.31 |

Data are presented as n (%), mean±SD or median (interquartile range). NA: not available. Bold values indicate significant p-values.

### *Stable* versus *exacerbating participants*

The mean HR and RR were numerically higher in the exacerbating participants compared with the stable participants. For the HR time-series, the exacerbating population had significantly higher variability measures with a median cSDNN (268 ms (207–315) *versus* 179 ms (162–246); p=0.006), median SD1 HR (12 ms (8–15) *versus* 8 ms (7–10); p=0.04), median SD2 HR (90 ms (79–130) *versus* 75 ms (67–96); p=0.037), but had a significantly lower complexity measure with a median SEHR (0.1258 (0.1044–0.1568) *versus* 0.1607 (0.1355–0.1981); p=0.015) compared with the stable population (table 4).

For the RR time-series, the exacerbating population had a significantly higher variability and complexity measures with a median SDBB (629 ms (515–1126) *versus* 525 ms (398–665); p=0.024), median SD1 RR (134 ms (96–169) *versus* 92 ms (76–109 ms); p=0.037) and median SD2 RR (876 ms (722–1584) *versus* 767 ms (558–936); p=0.025) compared with the stable population (table 3).

For the airflow time-series, both populations demonstrated two crossover points in the DFA which were at similar time points leading to three scaling exponents (supplementary figure S1). The exacerbating population had a significantly lower median α1 (1.63 (1.61–1.65) *versus* 1.66 (1.63–1.67); p=0.005) and median α3 (0.50 (0.47–0.56) *versus* 0.69 (0.65–0.79); p <0.001) compared with the stable population (table 3).

MSE analysis of both HR and RR showed that the SE significantly increased as the scale increased for both the stable and exacerbating groups. A two-way ANOVA test showed a significant reduction in HR MSE ($F_{group}$ (1,20)=17.895; p<0.001, $F_{scale}$ (9,20)=38.37; p<0.001) in the exacerbating group compared with the stable group, irrespective of scale. While there was a significant increase in the RR MSE ($F_{group}$ (1,20)=40.703; p<0.001, $F_{scale}$ (9,20)=51.434; p<0.001) in the exacerbating group compared with the stable group (figures 2 and 3).

For the stable COPD group, there were no significant correlations between $FEV_1$, CAAT score, mMRC score and any of the time-series variability measures. In the exacerbating group, there was no significant correlations found between any of the time-series variability measures and the admission NEWS2, the C-reactive protein or the Rome COPD exacerbation severity classification.

**TABLE 3** Pulmonary function tests, baseline symptom severity assessment scores and relevant medication of stable *versus* exacerbating participants

| Characteristic | Stable group | Exacerbating group | p-value |
|---|---|---|---|
| **Pulmonary function tests**[#] | | | |
| $FEV_1$, L | 1.35±0.64 | 0.91±3.22 | **0.008** |
| $FEV_1$ % predicted | 48.96±20.00 | 32.31±10.14 | **0.002** |
| FVC, L | 3.27±1.06 | 2.42±0.68 | **0.004** |
| FVC % predicted | 91.31±25.67 | 70.00±19.97 | **0.005** |
| $FEV_1$/FVC ratio | 0.41±0.12 | 0.38±0.11 | 0.47 |
| $T_{LCO}$, mmol·min$^{-1}$·kPa$^{-1}$ | 4.11±2.34 | 2.46±0.63 | **0.023** |
| $T_{LCO}$ % predicted | 48.60±22.82 | 33.21±10.78 | **0.029** |
| $K_{CO}$, mmol·min$^{-1}$·kPa$^{-1}$ | 0.86±0.35 | 0.65±0.18 | 0.06 |
| $K_{CO}$ % predicted | 61.88±25.52 | 48.72±16.02 | 0.10 |
| **Symptom assessment questionnaires** | | | |
| mMRC breathlessness score | 3 (2–3) | 3 (2.5–4) | 0.25 |
| CAAT score | 20 (14–25.25) | 27 (22.5–32.25) | **0.004** |
| **COPD severity assessment (GOLD)** | | | |
| GOLD A | 2 (6) | 0 | **0.003** |
| GOLD B | 20 (61) | 3 (17) | **0.003** |
| GOLD E | 11 (33) | 15 (83) | **0.003** |
| **Respiratory medication** | | | |
| SABA | 33 (100) | 18 (100) | NA |
| LABA/ICS | 0 | 1 (6) | 0.17 |
| LAMA alone | 0 | 1 (6) | 0.17 |
| LABA/LAMA[¶] | 10 (30) | 1 (6) | **0.040** |
| LABA/LAMA/ICS[¶] | 21 (64) | 15 (83) | 0.14 |

Data are presented as mean±SD, median (interquartile range) and n (%). $FEV_1$: forced expiratory volume in 1 s; FVC: forced vital capacity; $T_{LCO}$: transfer factor of the lung for carbon monoxide; $K_{CO}$: transfer coefficient of the lung for carbon monoxide; mMRC: modified Medical Research Council; CAAT: COPD and Airways Assessment Test; GOLD: Global Initiative for Chronic Obstructive Lung Disease; SABA: short-acting β agonist; LABA: long-acting β agonist; LAMA: long-acting muscarinic antagonist; ICS: inhaled corticosteroids; NA: not available. [#]: All participants had lung function tests within 3 years of the study. In some cases, the lung function test after the study start was taken as it was closest. [¶]: Included patients on various combinations, but receiving all medications. Bold values indicate significant p-values.

**TABLE 4** Differences in physiological signals comparing the stable COPD group with the exacerbating group

| Physiological variability measure | Stable COPD groups | Exacerbating COPD group | p-value |
|---|---|---|---|
| **HR measures (n=31 *versus* 9)** | | | |
| Mean HR, bpm | 71.04 (64.55–75.10) | 74.22 (67.42–82.31) | 0.35 |
| cSDNN, ms | 179.40 (162.14–245.83) | 268.41 (206.67–314.90) | **0.006** |
| SD1 HR, ms | 7.80 (6.52–10.40) | 12.06 (7.76–14.51) | **0.040** |
| SD2 HR, ms | 74.76 (66.68–95.67) | 90.18 (78.62–130.15) | **0.037** |
| SEHR | 0.1607 (0.1355–0.1981) | 0.1258 (0.1044–0.1568) | **0.015** |
| **RR measures (n=32 *versus* 18)** | | | |
| Mean RR, breaths·min$^{-1}$ | 16.86 (15.13–18.87) | 18.58 (15.27–20.50) | 0.30 |
| SDBB, ms | 525.41 (398.08–665.35) | 628.68 (514.90–1126.30) | **0.024** |
| SD1 RR, ms | 91.97 (75.97–108.82) | 133.61 (95.52–168.53) | **0.037** |
| SD2 RR, ms | 737.48 (558.01–936.16) | 875.93 (722.42–1584.36) | **0.025** |
| SERR | 0.1078 (0.0743–0.1377) | 0.1378 (0.0982–0.1916) | 0.09 |
| **Airflow analysis (detrended fluctuation analysis) (n=29 *versus* 12)** | | | |
| α1 | 1.6607 (1.6341–1.6730) | 1.6294 (1.6055–1.6452) | **0.005** |
| α2 | 0.2805 (0.2600–0.3077) | 0.2968 (0.2779–0.3110) | 0.13 |
| α3 | 0.6931 (0.6509–0.7931) | 0.5000 (0.4676–0.5578) | **<0.001** |

Data are presented as median (interquartile range). HR: heart rate; bpm: beats per min; SDNN: the SD of the inter-beat interval of successive heart beats; cSDNN: the SDNN corrected for mean HR; SD1 HR: measure of short-term HR variability; SD2 HR: measure of longer-term HR variability; SEHR: measure of signal HR complexity and richness; RR: respiratory rate; SDBB: SD of the breath-to-breath interval; SD1 RR: measure of short-term RR variability; SD2 RR: measure of longer-term RR variability; SERR: measure of RR signal complexity and richness. Mann–Whitney U-test was performed comparing the stable COPD and exacerbating COPD population. Bold values indicate significant p-values.

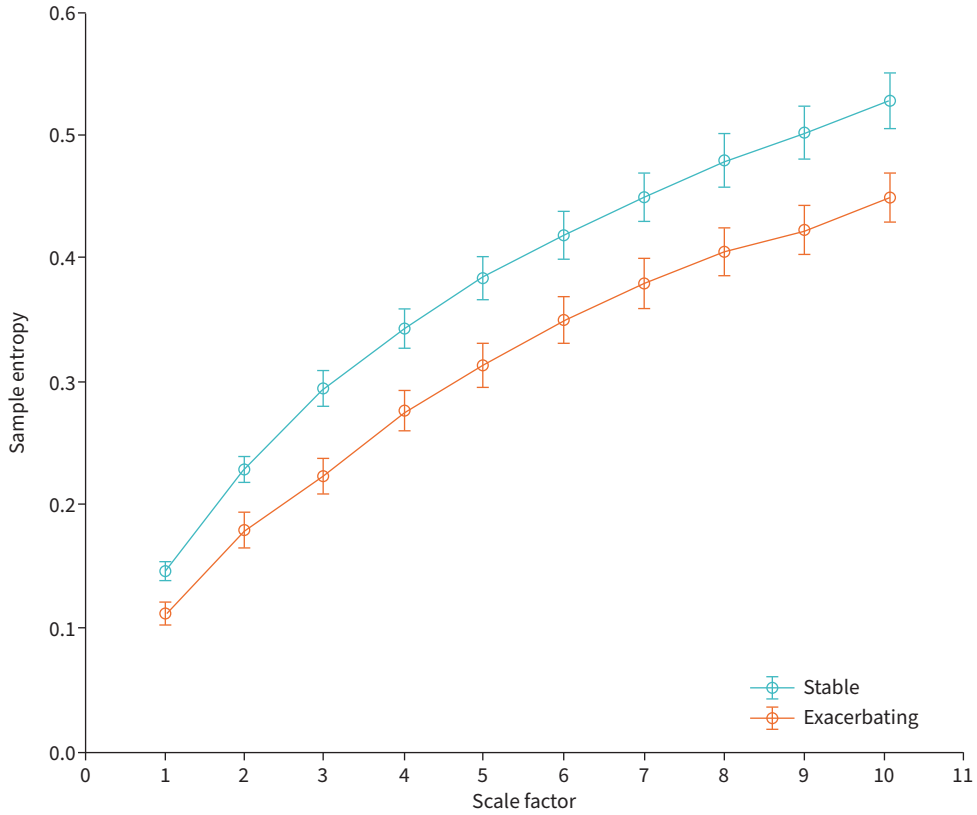


FIGURE 2 Heart rate multiscale entropy (MSE) comparing stable COPD participants and exacerbating participants

### *Exacerbating participants: admission versus post-discharge*

For the exacerbating group, nocturnal data 5 days after discharge data were available in 9 of 18 (50%) of participants. Given the small numbers, statistical analysis with a Friedman's test was not computed. At 5 days after discharge, the mean HR was largely unchanged, but the variability measures (cSDNN, SD1 HR and SD2 HR) trended downwards, whereas SE of HR increased. While the mean RR was also largely unchanged, the other variability measures all decreased at 5 days after discharge compared with admission values. This is illustrated in supplementary figures S4 and S5. Airflow measures α1 and α2 decreased from admission to post-discharge, whereas α3 increased. This is illustrated in supplementary figure S6.

For HR, MSE increased 5 days after discharge at all scales (supplementary figure S7). For RR, multiscale entropy decreased 5 days after discharge at all scales (supplementary figure S8).

Usability of AcuPebble in the stable COPD population was obtained from 24 of 33 (73%) of participants with the majority (16 of 24) finding attaching the sensor very easy. Most participants (17 of 24) found the sensor comfortable to wear and in 20 of 24 the sensor stayed in place for the duration of the night. Usability feedback was obtained from 8 of 18 participants in the exacerbating group. All participants found the sensor at least moderately easy to put on, with 75% finding it comfortable. Only one participant developed a rash around the site of the device; however, this disappeared through the day.

The results from our historical non-COPD group, using a similar device can be seen in the supplementary material (pp. 16–20). The non-COPD group were younger with fewer comorbidities. No significant differences were found in the HR and RR variability measures between the non-COPD group and stable COPD group. There was a significant difference in airflow variability measures with α2 being significant lower in the non-COPD group *versus* stable COPD participants (0.2367 (0.2070–0.2661) *versus* 0.2805 (0.2600–0.3077); $p<0.05$). HR MSE was higher in the non-COPD group compared with the stable and exacerbating COPD groups, while RR MSE was lower in the non-COPD groups compared with both COPD groups.

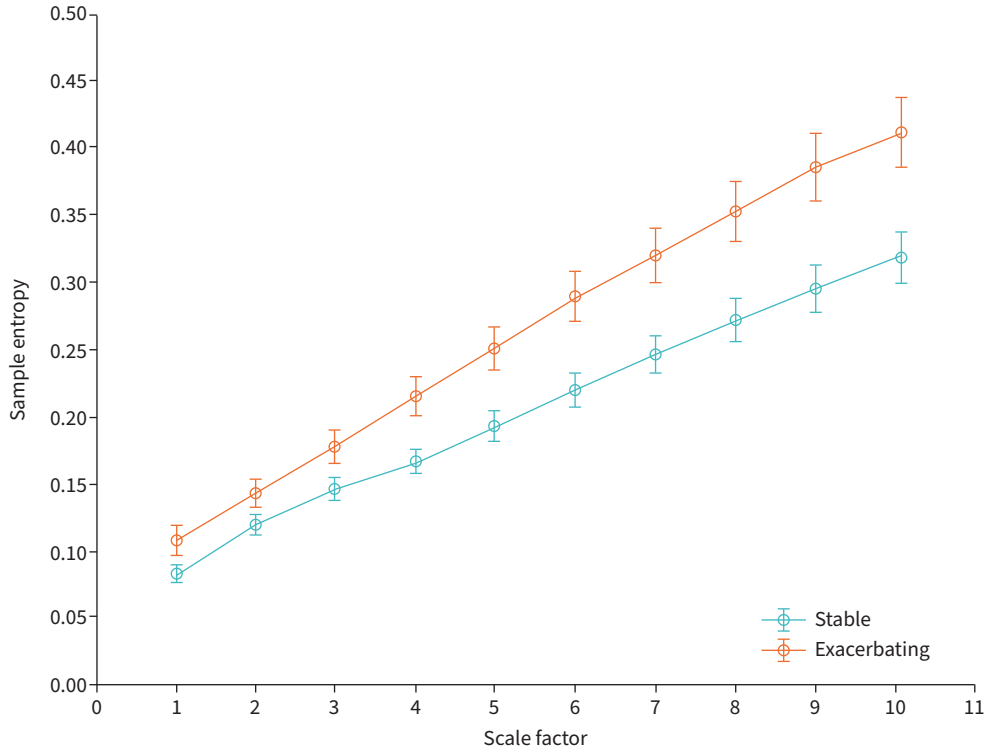


FIGURE 3 Respiratory rate multiscale entropy (MSE) comparing stable COPD participants and exacerbating participants

## Discussion

This prospective observational study has shown that significant differences exist in HR and RR variability measures between stable COPD participants and those undergoing an exacerbation. The differences are summarised in figure 4.

HR variability (HRV), measured by cSDNN, SD1 and SD2, was significantly increased in participants undergoing an exacerbation compared with the stable population. These findings are similar to previously

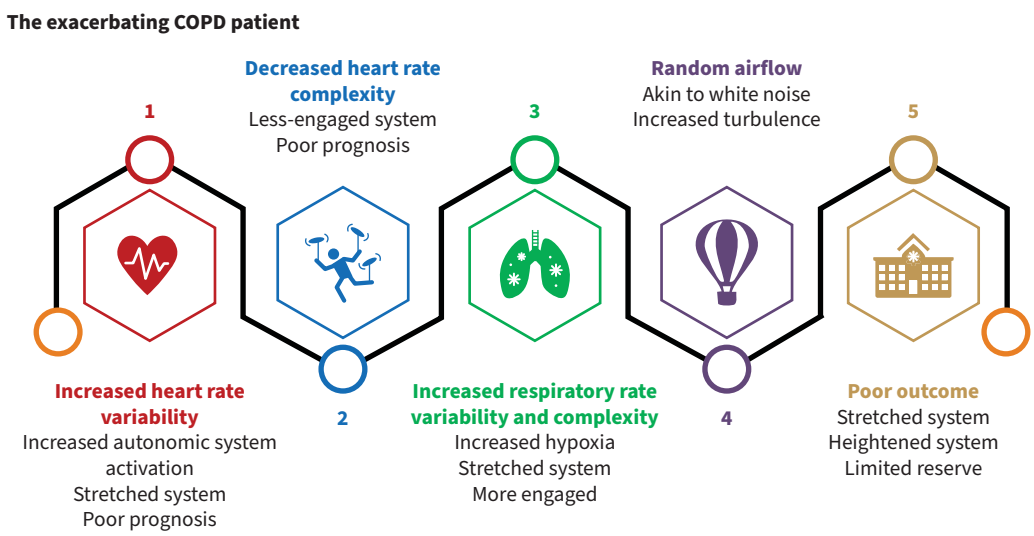


FIGURE 4 Variability analysis summary

published data by Kabbach *et al.* [17], who used 10-min daytime recordings comparing COPD exacerbators with stable participants. They also found increased HRV measures (increased SDNN, SD1 and SD2) in the exacerbating group. Another study also found similar results using spectral analyses, assessing the HRV using frequency domains, showing increased variability in the exacerbating population [18]. We did not find a statistically significant increase in mean HR in the exacerbating participants, which was 3 bpm higher compared with the stable population. Previous work has suggested that daytime HR typically rises by about 7 bpm at exacerbation [19]. The nocturnal measurements used in this study show a smaller change.

HR is controlled by the autonomous nervous system (ANS) with a complex interplay between the parasympathetic nervous system (PNS) and the sympathetic nervous system (SNS). The PNS primarily impacts beat-to-beat variability [10], and this is reflected in short-term measures of HRV such as SD1 HR. We found that SD1 HR significantly increased in the exacerbating population, suggestive of increased short-term variability. This is likely to be due to increased respiratory sinus arrhythmia and possibly increased PNS activation. In humans, airway tone is thought to be mainly vagally controlled [20] and during an exacerbation, the increased bronchoconstriction, airway narrowing and increased vagal activity in the airway, may translate into increased PNS activity [17].

Global ANS activity is reflected in longer-term measures of HRV such as cSDNN and SD2 HR [10]. We found that these longer-term measures were significantly higher in the exacerbating group *versus* stable group (cSDNN 268 ms (207–315) *versus* 179 ms (162–246); p=0.006) suggestive of increased ANS activation, which is likely to be due to both increased PNS activity as described above, but probable also increased SNS activity due to the use of β-agonists and infective/inflammatory elements that play a role in the exacerbation.

It is important to note, that while in health a higher HR variability is linked to better prognosis, this is unlikely to be the case in this situation, as increased variability relates to a stretched and heightened ANS and worse outcomes [17].

SE quantifies the degree of irregularity *versus* regularity in a time-series and measures complexity and structural richness of the signal. Low values, suggest decreased complexity, more regularity and less system engagement. It takes into account the multiple regulatory systems affecting a particular time-series [21–23]. SE HR is significantly lower in participants undergoing an exacerbation, suggesting increased regularity and less complexity. MSE further defines the richness/complexity of the time-series by considering the multiple timescales that exist in physiological systems. MSE analysis of HR for both groups increased, confirming that HR is a complex time-series [13, 24, 25]. Participants with an exacerbation had decreased MSE HR at all timescales. Previous work has shown lower SE of HR in septic patients [26] and cirrhotic patients [11] and that HR MSE is an independent predictor of death in patients hospitalised for trauma [27]. A reduction in entropy describes a reduction in system coupling, a less-engaged system and one that cannot adapt to added stressors [28, 29].

RR can be affected by a myriad of different signalling pathways and a voluntary component differs in the awake and sleep state. Our work is novel, with no previous study investigating RR variability in participants undergoing a COPD exacerbation. The mean RR rose in the exacerbating population by a similar level compared with previous work [19].

RR variability measures (SDBB, SD1 RR and SD2 RR) were significantly higher in participants having an exacerbation. MSE analysis of RR showed that SE was also significantly higher in the exacerbating group compared with stable participants. Furthermore, the difference was more apparent at higher scales, suggesting that the increased signal complexity was more apparent when longer-term timescales were considered. This is useful in clinical practice as it suggests that wearables with a lower resolution (*e.g.* one sample every 10 or 20 s) can still provide valuable information. This increased variability and complexity can be explained in a few ways. Participants undergoing an exacerbation are more hypoxic, leading to metabolic imbalances and increased RR, coupling and engagement. During a COPD exacerbation there is greater skeletal muscle dysfunction [30] and increased hyperinflation [31], meaning reduced efficacy of both the inspiratory muscles and diaphragm. This leads to a fall in minute ventilation and increased engagement from the respiratory system with increased complexity.

No previous study has looked at nocturnal airflow in this population. We have shown that COPD participants have two crossover points in DFA analysis and three exponents (α1, α2 and α3). This indicates the scaling exponent (α) differs for different range of scales. While there was no clinically

relevant difference in α1 and α2, at longer scales (above 30 s), exacerbators had a significantly lower α3 value compared with stable participants (median (IQR) 0.5000 (0.4676–05578) *versus* 0.6931 (0.6509–0.7931); p<0.001). This suggests that while stable COPD participants demonstrate long-range power law dynamics and positive autocorrelation at longer scales (whereby higher airflow is followed by higher airflow and vice versa), the exacerbating population have an α3 of 0.5, which is suggestive of white noise and completely random fluctuation. This is likely explained by increased upper and lower airway inflammation as well as bronchoconstriction, oedema and increased mucous production. This can lead to expiratory flow limitation, narrow airway calibre and increased turbulence of flow [31].

Of note, for all the physiological time-series, there was a trend towards "normality" or stable COPD participant values in the exacerbating participants by day five after discharge. This is important as it suggests that in the recovery phase of COPD, the variability measures return to baseline.

There are strengths and limitations to consider to this work. Participants were representative of a moderate–severe COPD population and were symptomatic, with most participants already on triple inhaled therapy. There were no significant differences in age, sex, smoking history and medical comorbidities between the groups. The exacerbation cohort had significantly worse historic lung function with a reduced $FEV_1$, FVC and transfer factor of the lung for carbon monoxide and a higher CAAT symptom score on admission. While we used the median nocturnal time as the start point of our analysis for the exacerbating population, no sleep diaries were given to patients. Therefore, we may have captured some data while COPD exacerbators were awake. This could lead to some bias, as there is evidence to suggest HR and RR variability differs during sleep compared with wakefulness. Second, there was a degree of artefact in the HR and airflow recordings, meaning a small sample size and therefore a high type 2 (β) error. Third, the patients admitted to hospital may have started their actual exacerbation at different times. In future studies, a retrospective symptom diary would be useful to pinpoint the exact start of their symptom onset. Fourth, data capture both preceding the exacerbation and post-exacerbation would have been useful to map the exacerbation trajectory. Fifth, both BMI and lung function can potentially influence the autonomic tone, thus impacting variability analysis. However, given the small numbers, multivariable analysis could not be performed. Moreover, from the stable group, no correlations were seen between any of the time-series parameters and $FEV_1$. This suggests that differences in lung function between stable and exacerbating states are unlikely to have contributed to the differences seen.

Overall, we have shown that continuous measurements of HR, RR and airflow through wearable technology is feasible and acceptable to patients during states of stability and exacerbation. There are clear differences in time-series variability and complexity measurements between the stable and exacerbating COPD populations (figure 4). Therefore, understanding and integrating variability measures into clinical practice is important to be able to build exacerbation detection algorithms in the future. Future work needs to focus on identifying the point(s) at which variability measures change in a patients' exacerbation journey and whether this can be picked up objectively prior to symptoms. This will help build algorithms in the future such that treatment can be started earlier and thus avoid severe sequalae. Signal changes will also be useful for monitoring recovery and identifying those at risk of recurrent exacerbation and readmission.

The role of continuous physiological measurement, while in its infancy, has great potential to change the detection and management of exacerbations, leading to improved patient outcomes in the future.

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Author contributions: E. Rodriguez Villegas and S. Mandal developed the original methodology for the study together. A.J. Shah recruited participants, collected data, analysed data and performed the linear and nonlinear analysis with A.R. Mani. O. Dessi helped with trial and device management. A.J. Shah wrote the initial draft of the paper. A. Saigal helped with data collection and data analysis. R. Pramono created software to facilitate extraction of the datapoints in a format that facilitate subsequent processing. A.J. Shah, S. Mani, A.R. Mani and J.R. Hurst made substantial contributions to the interpretation of the data. All authors reviewed and revised the manuscript.

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